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How education reduces urban relative poverty? Evidence from China

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ABSTRACT

Previous studies mainly focus on absolute poverty in China. However, the rate of relative poverty of Chinese urban residents has reached 23.90% in 2018. This paper investigates the mechanism for reducing urban relative poverty through education. Findings are that the increase of family education years can help urban households get rid of relative poverty through the indirect way of improving social capital. Further study finds that greater influence of Confucian culture and weaker marketization degree will result in stronger mediating role of social capital between education and poverty reduction.

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1. Introduction

Although at the end of 2020, the absolute poverty in China had been eliminated under its official standard, the problem of relative poverty is still serious. According to data from the Ministry of Civil Affairs of China, as of 2021, China has identified as many as 58 million relatively poor people, including 4.31 million people living on the edge of basic living allowances. Furthermore, the potential benefits of urbanization are not generalized among the urban poor households. As of 2021, the urbanization rate of China has reached 64.72%, which means urban population has exceeded the rural population. However, urban relative poverty is more serious than rural relative poverty. Based on the China Family Panel Studies 2018 (CFPS2018), it can be calculated that the relative poverty incidence rate of Chinese urban residents was 23.90%. Paying closer attention to the problem of relative poverty in Chinese cities is a priority for researches, policies and social programs.

As another poverty classification standard, relative poverty refers to the situation in which one can maintain a basic life by relying on one's own ability or assistance from the government and society, but the living standard is still low, which is lower than the normal living standard of local residents. Economic growth has brought about an improvement in the average living standard, but at the same time, it has also brought about a widening of the income gap, and the problem of relative poverty caused by this has become more and

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more obvious. Compared with poorer rural areas, the incidence of relative poverty in urban areas is higher (Walker and Yang 2020). Relatively poor households in cities are not only in material difficulties, but also lack development opportunities and capabilities. This is obviously unsustainable, so how to improve the development capacity of the relatively poor urban population through education is an urgent problem to be studied.

The poverty reduction effect of education has long been concerned by scholars, and existing researches focus more on the poverty reduction effect of education by increasing human capital. From a macro point of view, the government's fiscal expenditure on education can improve the human capital of the offspring of poor families, thereby narrowing the income gap between the offspring of families with different incomes and having a very significant effect on poverty reduction (Solon 2004). From a micro perspective, compared with direct relief measures, education can not only help poor families improve labor productivity, but also improve their health, social integration, and development motivation, thereby getting rid of their dependence on relief (Wolfe and Zuvekas 2000). Therefore, the anti-poverty effect of education is more durable. Among various types of education, compulsory education has the strongest poverty reduction effect (Stephens Jr and Yang 2014). Among non-compulsory education, vocational education is stronger than general education (Kahyarara and Teal 2008).

Existing researches have achieved a lot of results, but there are still the following shortcomings. Firstly, previous studies have focused more on the role of education in poverty reduction by increasing human capital, and less attention has been paid to the mediating role played by social capital (Woessmann 2016). Secondly, the existing literature pays more attention to the heterogeneity of poverty reduction in vocational education and compulsory education, and basically does not pay attention to the heterogeneity caused by the differences of degree of Confucian cultural influence and marketization (Kahyarara and Teal 2008; Stephens Jr and Yang 2014). Thirdly, the existing literature focuses more on absolute poverty than relative poverty (Walker and Yang 2020). Fourthly, the discussion on urban relative poverty is far less popular than the discussion on rural relative poverty (Gustafsson and Sai 2020).

Based on CFPS2018 data, this paper empirically studies the influence mechanism and regional heterogeneity of education on urban relative poverty. The possible marginal contributions are as follows: Firstly, the mediation effect model is used to deeply analyze the mediating role of social capital in poverty reduction through education. Innovations are made in the research perspective. Secondly, it further analyzes the regional heterogeneity of the mediating role of social capital under different cultural backgrounds (Confucian culture) and marketization degrees, providing a deeper understanding of the mechanism of education in poverty reduction through social capital. Thirdly, under the background that the problem of relative poverty in Chinese cities is a priority, it is more practical to expand the research on education's role in poverty reduction to the field of urban relative poverty.

2. Literature review

2.1. Education and urban relative poverty

Research on the relationship between education and poverty has a long history. From a macro perspective, the increase in fiscal expenditure in education will improve a

country's education level, thereby significantly eliminating poverty. From a micro level, modern human capital theory regards education as a productive investment behavior. Therefore, education can increase the accumulation of human capital, improve labor productivity, enable the poor to obtain more and better job opportunities, and increase the income of poor families, thereby reducing the incidence of poverty.

The existing literature pays more attention to the effect and mechanism of education in alleviating relative poverty in rural areas. López and Valdés (2000) studied the panel data of six Latin American countries and found that the contribution of one additional year of education on rural households' annual income per capita in five of the six countries ranges from \$3.70 to \$62.00 (López and Valdés 2000). Liu et al. (2023) used a spatial autoregressive model (SAR) to study the spatial agglomeration effect of rural education level and rural poverty in 27 provinces in China from 2010 to 2017. The results of the study show that different education levels have a significant impact on rural poverty, and the impact of compulsory education on rural poverty is greater than that of high school education. Further research finds that rural poverty in a region is clearly not only driven internally, but also affected by poverty rates in surrounding provinces (Liu, Li, and Zhao 2023).

In recent years, there have also been empirical studies on the impact of education and other factors on urban relative poverty, but mainly from the perspective of specific types of poor population, such as floating population and migrant workers, and there are few studies on the mechanism of education alleviating relative poverty. Chen and Zhi (2021) studied the contribution rate of education, housing and other factors to alleviating the relative poverty of the floating population based on the Chinese floating population dynamic monitoring data in 2018 (Chen and Zhi 2021). Their research results show that the poverty reduction effect of education on the floating population is most significant in the Pearl River Delta, the Yangtze River Delta, the middle reaches of the Yangtze River and the Chengdu-Chongqing urban agglomeration, and the poverty reduction contribution rate in the Beijing-Tianjin-Hebei urban agglomeration is second only to housing.

Based on the abovementioned literature analysis, this paper believes that education can significantly improve the relative poverty problem in cities. This poverty reduction effect is mainly manifested in two aspects. Firstly, education can increase the accumulation of human capital, improve the labor productivity of the relatively poor population, and substantially make them possess the ability to perform difficult jobs. Secondly, according to the signaling theory, employers will identify job applicants' abilities by their academic qualifications. Therefore, the increase in the number of years of education of relatively poor people can transmit stronger ability signals to the job market, and thus may have better job opportunities.

Based on the above analysis, this paper proposes the following assumption:

Hypothesis 1: Education can significantly reduce the incidence of relative poverty in urban households.

2.2. Mediating effect of social capital between education and urban relative poverty reduction

Social capital is the sum of actual or potential resources in a social network (Putnam and Leonardi 1994), and a social network consists of familiar relationships, among

which the relationships between classmates and teachers and students are the most important relationships. As an investment behavior, education not only produces human capital, but also social capital. Primary school, junior high school, high school, and college classmates are often a person's most reliable and intimate connections. In addition, when graduating, teachers will use their own life experience to help students eliminate unreliable jobs, and some university teachers with sufficient social resources will even help introduce jobs. Education can also bring about an alumni identity and make it easier to develop mutual trust, especially for students who graduated from the same university (Rust and Uys 2014). Glaeser et al. (1999) analyzed data from the World Values Survey and claimed a positive effect of education on social capital. Huang (2009) provided evidence that education is a strong and robust correlate with individual social capital.

Based on the above analysis, this paper proposes the following assumption:

Hypothesis 2: The increase in the number of years of education can significantly increase the social capital of the family.

In terms of research of social capital alleviating absolute poverty, Zhang et al. (2007) studied the impact of social capital on absolute poverty in rural areas, and the results showed that social networks can significantly reduce poverty (Zhang, Lu, and Zhang 2007). Furthermore, the poverty reduction effect of social capital at the family level will decrease with the increase of the degree of marketization, while the poverty reduction effect of social capital at the social level will not significantly reduce. Through mathematical modeling and data simulation, Chantarat and Barrett (2012) found that, in the case of insufficient physical capital, social capital can replace physical capital to improve the productivity of absolute poor households, thereby increasing their income level (Chantarat and Barrett 2012). The increase in income can in turn cultivate the family's social skills, thereby accumulating more social capital. Absolutely poor families can escape the 'poverty trap' through such a virtuous cycle. Guo et al. (2014) constructed a lineage network model and found that social capital related to lineage network can play a role in alleviating absolute poverty in rural areas (Guo, Yao, and Jeremy 2014).

In terms of research on social capital reducing relative poverty level, research by Grootaert (2002) found that social capital can increase employment opportunities, increase the income of poor groups, narrow income gaps and resist risk shocks, and it has an immeasurable role in reducing the incidence of poverty (Grootaert, Oh, and Swamy 2002). Zhang et al. (2017) analyzed the impact of social capital on residents' poverty, and the results show that social capital can alleviate the income gap, thereby reducing the possibility of residents falling into relative poverty (Zhang, Zhou, and Lei 2017).

Based on the abovementioned literature analysis, this paper argues that an increase in social capital accumulation can improve urban relative poverty. This influence is manifested in two aspects. Firstly, social capital can replace physical capital to a certain extent, because personal connections are also a kind of resource that can provide financial security and emotional support for the relatively poor. How can social capital replace material capital? For example, poor urban residents who set up stalls to do small businesses may be fined by the Urban Management Bureau for violating urban management regulations. However, if he knows the civil servants of the Urban Management Bureau, he will know how to avoid violating the regulations and will not

be fined. Potential fines are material capital, while being friends of the civil servants of the Urban Management Bureau is social capital. The social capital owned by poor residents helps him avoid the loss of material capital. People are willing to spend money to maintain social relations, because social relations can bring them more gains, or at least avoid potential losses. Secondly, the increase of social capital can reduce the asymmetry of employment information, so that relatively poor people have the opportunity to obtain better job opportunities. In addition, social capital can also provide initial financial support for self-employment. Technical secondary school teachers and university teachers can help introduce jobs, or at least prevent students from entering unscrupulous enterprises. Students will also introduce jobs to each other, broadening the employment information channels of relatively poor urban residents.

Based on the above analysis, this paper proposes the following assumption:

Hypothesis 3: Education can indirectly improve urban relative poverty by increasing the accumulation of social capital.

The specific mechanism of education improving urban relative poverty directly and indirectly through social capital is shown in Figure 1.

2.3. Regional heterogeneity caused by Confucian culture and marketization

If education can increase the accumulation of social capital, and the increase in social capital can help urban households get rid of relative poverty, is there any heterogeneity in this mechanism among different regions?

Firstly, this paper argues that the greater the influence of Confucian culture, the stronger the role of education in poverty reduction through social capital. There are two reasons for this. Firstly, education increases the accumulation of social capital through the relationship between classmates, teachers and alumni, and the construction of interpersonal relationships is strongly influenced by Confucian culture. Confucian ethics regulates the behavior standards of monarchs and ministers, teachers and students, fathers and sons, and brothers, classmates, relatives and friends. Affected by this, in Chinese society, people use more identity norms rather than institutions to build trust relationships. Secondly, under the background of Confucian

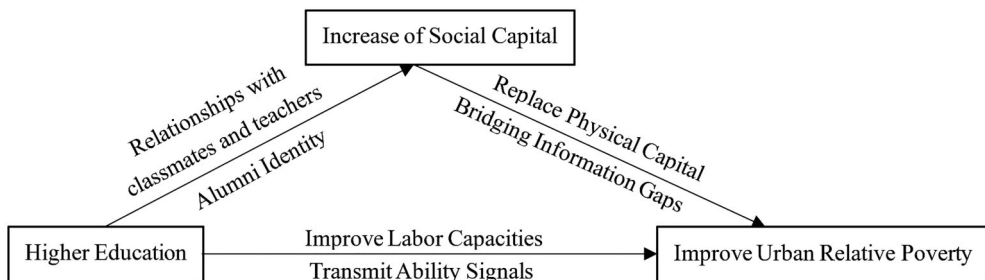


Figure 1. The Mechanism of Education Improving Urban Relative Poverty.

Education can not only directly improve urban relative poverty by improving labor capacities and transmitting ability signals, but also indirectly alleviate urban relative poverty through increasing social capital. The specific mechanism is that education increases social capital through teacher, classmate and alumni relationships, which can replace material capital and bridge information gaps, thereby reducing urban relative poverty.

culture, economic transactions are more often achieved by blood relationship or circle of acquaintances, and the cost of reaching transactions between strangers is very high (Du 2015). In this case, for urban households, employment opportunities are more likely to be obtained through introductions from acquaintances. If a wide enough social network is not created through education or other means, they are likely to fall into relative poverty. Based on the above analysis, this paper proposes the following assumption:

Hypothesis 4a: The greater the influence of Confucian culture, the stronger the poverty reduction effect of education through social capital.

Secondly, this paper argues that the higher the degree of marketization, the weaker the role of education in poverty reduction through social capital. Social capital can form informal institutions and effectively compensate for market defects (Bowles and Gintis 2002). As mentioned earlier, an increase in social capital accumulation can bridge the information gap in the job market. In addition, in areas with immature market mechanisms, the more social resources one person has, the more physical capital they can possess, and social capital can replace physical capital to some extent. However, in regions with strong market forces, information gaps can be closed, and those who enjoyed elite status by having more connections in the old economy also lose their advantages (Nee 1991). The role of social capital will be weakened by market forces. Therefore, in areas with a higher degree of marketization, the role of social capital in poverty reduction will be weaker.

Hypothesis 4b: The higher the marketization degree, the weaker the poverty reduction effect of education through social capital.

3. Research methodology and data

3.1. Data sources

The data set used in this article is the China Family Panel Studies 2018 (CFPS2018) released by the Institute of Social Science Survey (ISSS) of Peking University, China. It surveyed 15,000 households in 31 provinces of China and collected 44,000 individual questionnaires¹. At the family-level, CFPS2018 mainly investigates family income and expenditure, property conditions, and living conditions, while the individual-level investigation mainly involves the education level, depression level, health status, and cognitive level of family members. The samples of CFPS2018 are distributed in nearly 3,000 villages in more than 900 districts and counties, representing 95% of China's population.

This paper studies urban relative poverty, so the samples whose residence is in the countryside should be excluded, and then the samples that lack important information are excluded. After screening, the final samples include 6,594 urban households.

3.2. Variables

3.2.1. Measurement of urban relative poverty

At present, there is no consensus among scholars about the threshold for relative poverty. There are three main methods for identifying relative poverty. The first is

called ‘strongly relative poverty lines’ which are set at 40% to 60% of the mean or median income. For example, Gottlieb and Fruman (2011) regarded 60% of the median annual household income per capita as urban relative poverty line (Gottlieb and Fruman 2011). The second is using the Alkire-Foster method to construct multi-dimensional poverty index. For example, Alkire and Fang (2019) constructed a multi-dimensional poverty index in China and compared it with income poverty using multi-wave panel data from the China Health and Nutrition Survey (Alkire and Fang 2019). The third is the weakly relative poverty line proposed by Ravallion and Chen (2011) (Ravallion and Chen 2011).

This paper will use the weakly relative poverty line as the threshold of urban relative poverty. There are two reasons. Firstly, compared with the strongly relative poverty lines, weakly relative poverty lines take the cost of social inclusion into account. Secondly, this method simplifies the complex calculation of the second method.

In 2011, Ravallion and Chen proposed the weakly relative poverty line as follows (Ravallion and Chen 2011):

$$Z = \max(Z^*, \alpha + 0.5\bar{I}) \quad (1)$$

where Z^* represents the absolute poverty line, α is the lower bound of the social inclusion cost interval, and $\bar{I}$ is the mean or median annual household income per capita. The weakly relative poverty line Z can be calculated based on Equation (1). In 2017, they choose $Z = \max(\$1.9, \$0.4 + 0.5\bar{I})$ as the USD-a-day weakly relative poverty line at 2011 PPP after continuous optimization (Ravallion and Chen 2017). Based on Equation (1), this paper will calculate the CNY-a-day weakly relative poverty line in Chinese cities at 2018 PPP.

First of all, the official absolute poverty line of Chinese rural households obtained by the National Bureau of Statistics of China is 2,300 RMB. According to Ferreira et al. (2016), the line was 28.5% higher for urban compared to rural areas in China (Ferreira et al. 2016). Therefore, the absolute poverty line of Chinese urban households is 3,258 RMB per person per year in 2011 price, which is 3,767 RMB per person per year when converted into 2018 price.

Secondly, Ravallion and Chen (2017) estimated that the lowest social inclusion cost was US \$0.4 per person per day in 2011 price (Ravallion and Chen 2017). According to Ferreira et al. (2016), 1 USD per day in 2011 is equivalent to 3.9 CNY per day in Chinese cities, so the minimum cost of social inclusion in Chinese cities is 569.4 CNY per year at the 2011 price (Ferreira et al. 2016). Using each year’s CPI deflator, it is converted to 659.17 CNY per year at the 2018 price.

To sum up, this paper will calculate the CNY-a-day weakly relative poverty line in Chinese urban areas based on the formula $Z = \max(3767, 659.17 + 0.5\bar{I})$. The median annual urban household income per capita in the data set this paper used was 25,873.33 RMB, so the urban weakly relative poverty line should be 13,595.84 yuan. If the per capita annual income of a family in 2018 is less than or equal to 13,595.84 yuan, it is identified as relatively poor.

3.2.2. Measurement of education

The core explanatory variable studied in this paper is the household education level. Referring to the research of Cheng et al. (2014), it will be measured by the total years of education of family members over the age of 16 divided by the family size (Cheng et al. 2014).

3.2.3. Measurement of social capital

Before exploring the specific characteristics of social networks or investigating their quality, it is necessary to determine the type of network being studied in any given social capital study. Putnam (1998) distinguishes between informal and formal social networks, in which informal relationships include relationships within families, relatives, friends and neighbors, while formal relationships include relationships with social groups such as volunteer associations (Putnam 1998). How to measure social capital? Hofferth et al. (1995) proposed a way to measure the social capital of relatives: ask the respondents whether they will borrow money from relatives when they are in urgent need of money, and whether they have lent money to relatives (Hofferth, Boisjoly, and Duncan 1995). Baum et al. (2000) proposed a method to measure social capital of friend types: ask respondents how often they visit friends (Baum et al. 2000). Based on the above researches, this paper uses the family's money gift expenditure to measure social capital. In the robustness test, the frequency of communication between the family and relatives, friends, neighbors is used as the substitute variable of urban family's social capital.

It is appropriate to measure social capital with money gifts. Cao et al. (2022) and Su Jing (2019) have all adopted this measure of social capital (Cao et al. 2022; Su, Xiao and Hu 2019). Why can we use money gift expenditure to measure Chinese families' social capital? Since ancient times, gift giving has been one of the important means for Chinese families to maintain social relations. Unlike Western societies, where physical gifts are more common, Chinese families prefer to give monetary gifts. For example, at a wedding, guests need to give money to the bride and groom. During the Spring Festival, Chinese families will give money to the children of relatives and friends. This is called 'red envelope' in China (Hudik and Fang 2020). If social relations are regarded as a commodity with a fixed unit price, the higher the money paid to maintain social relations, the greater the number of social relations purchased. Specifically, the higher the expenditure, the more developed the social network and the higher the accumulation of social capital (Cao et al. 2022).

The values of social capital in data set are largely different from other variables and include zero, therefore this paper firstly adds 1 to the all values of social capital and then takes the logarithm.

3.2.4. Control variables

In order to scientifically set control variables, this paper selects control variables based on classical theories and literature.

Firstly, in order to overcome the heterogeneity of urban households, control variables should include household characteristic variables and household head characteristic variables. Drawing on the method of Ge et al. (2023), the 'financial respondent'

is identified as the head of the household (Ge et al. 2023). Due to the shortage of labor force in poor households, many urban relatively poor households have zero working-age population. Therefore, this paper uses the number of persons in the family as the denominator when calculating the household dependency ratio.

Secondly, in order to overcome the influence of regional heterogeneity, dummy variables for provinces are set to control region fixed effects and reflect factors related to urban residents' income, such as infrastructure and market maturity. The specific meanings of the selected explained variables, explanatory variables, mediator variables and control variables are shown in Table 1.

3.3. Empirical strategy

Firstly, examine Hypothesis 1, that is, the impact of education on urban relative poverty. The Probit regression model without social capital was constructed as follows:

$$Prob(Relapoverity = 1) = \Phi(\alpha_0 + \alpha_1 Edulen + \alpha_2 CV + \varepsilon_1) \quad (2)$$

where CV is the control variables in Table 1, and ε_1 is the error term.

Furthermore, the accumulation of social capital can replace physical capital to some extent and provide financial security and emotional support for the relatively poor. According to the above theoretical analysis, the increase of social capital can improve the relative poverty in cities. Therefore, social capital (*Socapital*) is introduced as a new control variable in Equation (2).

Table 1. Measurement of Variables.

Type of Variables	Variable Name	Measurement
Dependent Variable	<i>Relapoverity</i>	If the per capita annual income of a family in 2018 is less than or equal to the urban weakly relative poverty line, it is identified as relatively poor, i.e. <i>Relapoverity</i> = 1. Otherwise <i>Relapoverity</i> = 0.
Independent Variable	<i>Edulen</i>	It measures the household education level, which equals the total years of education of family members over the age of 16 divided by the family size.
Mediator Variable	<i>Socapital</i>	It measures the amount of social capital, which equals $\ln(1 + \text{expenditure of gifts})$.
Control Variables	<i>FS</i>	It measures the family size, which equals the number of persons in the family.
	<i>DR</i>	It measures the household dependency ratio, which equals $\frac{\text{family size} - \text{population of working age}}{\text{family size}}$.
	<i>FL</i>	It measures whether the family have borrowed money from banks or not: 1 = yes, 0 = no.
	<i>IL</i>	It measures whether the family have borrowed money from individuals and institutions other than bank or not: 1 = yes, 0 = no.
	<i>GENDER</i>	It identifies the householder's gender: 1 = male, 0 = female.
	<i>AGE</i>	It measures the householder's age.
	<i>MARRIED</i>	It identifies the householder's marital status: 1 = married, 0 = not married.
	<i>HEALTH</i>	It identifies the householder's health status: 1 = poor, 2 = fair, 3 = good, 4 = very good, 5 = excellent.
	<i>INS URANCE</i>	It identifies whether the householder has insurance or not: 1 = yes, 0 = no.
	<i>REGION</i>	Province dummies

$$Prob(Relapoverty = 1) = \Phi(\beta_0 + \beta_1 Edulen + \beta_2 Socapital + \beta_3 CV + \varepsilon_2) \quad (3)$$

Secondly, examine Hypothesis 2, that is, the impact of education on social capital accumulation. According to the theoretical analysis above, education can build relationships with classmates and teachers and bring about alumni identity, thereby increasing social capital. Therefore, Equation (4) is constructed as follows:

$$Socapital = \gamma_0 + \gamma_1 Edulen + \gamma_2 CV + \varepsilon_3 \quad (4)$$

Hypothesis 3 believes that social capital is the mediating variable of education alleviating urban relative poverty. According to the sequential test of mediating effect proposed by Baron and Kenny, the mediating effect of social capital is tested as follows (Baron and Kenny 1986).

Firstly, according to Equation (2), regress the explanatory variable Y ($Prob(Relapoverty = 1)$) on the explanatory variable X ($Edulen$) without the mediating variable M ($Socapital$). If the coefficient of X is not significant, the test will be terminated. If it is significant, regress Y on M and X as Equation (3), and regress M on X as Equation (4). If the coefficient of X is significant in formula (4) and the coefficient of M is significant in formula (3), then see whether the coefficient of X in formula (3) is significant. If it is significant, the result reflects a partial mediating effect. If it is not significant, the result reflects a complete mediating effect.

4. Empirical results

4.1. Education and social capital of relatively poor urban households

Table 2 below compares the years of education, family characteristics and householder characteristics of relatively poor families and non-relatively poor families. Relatively poor families account for about one-fifth of the sample, and their families have significantly fewer years of education than non-relatively poor families. At the same time, compared with relatively poor households, non-relatively poor households spend more on gifts for relatives, friends and neighbors, reflecting their more

Table 2. Comparison of relatively poor and non-relatively poor households.

Household's Type	Total		Relatively Poor		Non-relatively Poor	
	Mean	SD	Mean	SD	Mean	SD
Observations	6594	—	1576	—	5018	—
Proportion (%)	100.00	—	23.90	—	76.10	—
<i>Edulen</i>	8.67	4.30	6.06	3.66	9.48	4.16
<i>Socapital</i>	7.12	2.76	6.33	3.02	7.36	2.63
<i>FS</i>	2.92	1.60	3.43	1.82	2.76	1.49
<i>DR</i>	0.40	0.36	0.50	0.34	0.37	0.36
<i>FL</i>	0.09	0.28	0.07	0.25	0.09	0.29
<i>IL</i>	0.10	0.30	0.14	0.34	0.09	0.28
<i>GENDER</i>	0.49	0.50	0.48	0.50	0.50	0.50
<i>AGE</i>	49.08	15.50	53.02	14.41	47.83	15.63
<i>MARRIED</i>	0.81	0.39	0.82	0.38	0.81	0.39
<i>HEALTH</i>	2.89	1.14	2.67	1.23	2.96	1.09
<i>INSURANCE</i>	0.09	0.29	0.10	0.29	0.09	0.29

Notes: "SD" is the abbreviation of "Standard Deviation".

developed social network and higher social capital accumulation. The results preliminarily reflect the impact of educational level on the relative poverty status of urban families. The higher the educational level, the less likely it is to fall into relative poverty, and this effect can be achieved by increasing the accumulation of social capital.

4.2. Benchmark results

Table 3 shows the regression results of Equation (2), that is, the regression results of the relative poverty incidence on the years of education without introducing social capital. It shows that the years of education is significantly negative at the level of 1%, and the marginal effect on the incidence of relative poverty is -0.027^2 . The results indicate that education can improve urban relative poverty and Hypothesis 1 is established. According to modern human capital theory, education is a productive investment behavior that can increase the accumulation of human capital and substantially improve the labor capacity of the relatively poor population. In addition, because employers regard education as a signal of job seekers' ability, the increase in years of education can send a stronger signal of ability to the job market, which may bring better job opportunities. The regression results show that when other variables remain unchanged, the probability of urban households falling into relative poverty will decrease by 2.7% for each additional year of education.

Table 3. Benchmark Results.

Dependent Variable	<i>Prob(Relapoverty = 1)</i>	
	Coefficients	Marginal Effects
<i>Edulen</i>	−0.11*** (−22.16)	−0.03*** (−24.70)
<i>FS</i>	0.12*** (9.75)	0.03*** (10.01)
<i>DR</i>	0.27*** (3.98)	0.06*** (3.99)
<i>FL</i>	−0.18** (−2.44)	−0.04** (−2.44)
<i>IL</i>	0.13** (2.03)	0.03** (2.03)
<i>GENDER</i>	−0.10*** (−2.69)	−0.03*** (−2.70)
<i>AGE</i>	0.00*** (2.71)	0.00*** (2.72)
<i>MARRIED</i>	−0.14** (−2.54)	−0.03** (−2.55)
<i>HEALTH</i>	−0.10*** (−5.48)	−0.02*** (−5.52)
<i>INSURANCE</i>	0.20*** (2.95)	0.05*** (2.95)
Province Dummies	Included	Included
Constant	−0.980*** (−3.590)	—
Observations	6594	6594
Wald χ^2	1108.25*** (0.00)	—
Pseudo R^2	0.219	—

Notes: Robust t statistic in parentheses, while *, **, and *** indicate a significant level of 10%, 5%, and 1%, respectively.

Among the control variables of family characteristics, family size and dependency ratio are positively significant at the 1% level. This is also intuitive. If the number of family members is large and the family’s support burden is heavy, then the family is more likely to fall into relative poverty. Formal credit is negatively significant at the 5% level, and informal credit is positively significant at the 5% level. This is because households that are able to borrow from the banks usually have enough income or assets as collateral and are less prone to falling into relative poverty. Conversely, households that frequently borrow from relatives and friends are generally less resilient to risk and more likely to fall into poverty.

Among the control variables of householders’ characteristics, the gender and health status of the household head are negatively significant at the 1% level, and the age of the householder is positively significant at the 1% level. Firstly, when the head of the household is a woman, the householder is usually a single parent who needs to work and take care of children. The burden of living is heavy, and it is more likely to fall into poverty. Secondly, since the head of a household is usually the main labor force of the family, the worse his health is, the more likely the family is to fall into poverty. Thirdly, families with older householders are usually aging families, or families with heavier support burdens, which are less able to withstand risks and are more likely to fall into poverty. The marital status of the householder is negatively significant at the 5% level. One of the purposes of getting married is to enhance the ability to resist risks. Therefore, families with married heads are less likely to fall into relative poverty. The householder’s insurance is significantly positive at the 1% level, which is contrary to the common sense that purchasing insurance can improve the ability to resist risks. It is probably because too few households in the data set purchase insurance (only 9.16%). Another explanation may be that the more likely households are to fall into poverty, the more motivated they are to buy insurance.

4.3. Analysis of underlining mechanism

Table 4 shows the test results of the mediating effect of education on urban relative poverty through social capital.

Table 4. Results of the Mediating Effect Test.

Dependent Variable	Prob(Relapoverty = 1)		Socapital Coefficients	Prob(Relapoverty = 1)	
	Coefficients	MEs		Coefficients	MEs
<i>Edulen</i>	−0.113*** (−22.159)	−0.027*** (−24.695)	0.057*** (6.497)	−0.111*** (−21.582)	−0.026*** (−23.827)
<i>Socapital</i>	—	—	—	−0.086*** (−11.779)	−0.020*** (−12.186)
Control Variables	Included	Included	Included	Included	Included
Constant	−0.980*** (−3.590)	—	4.462*** (10.852)	−0.649** (−2.259)	—
Observations	6594	6594	6594	6594	6594
Wald χ^2	1108.25*** (0.00)	—	—	1154.03*** (0.00)	—
Pseudo R^2	0.219	—	—	0.238	—
R^2	—	—	0.123	—	—

Notes: (1) “MEs” is the abbreviation of “Marginal Effects”. (2) Robust *t* statistic in parentheses, while *, **, and *** indicate a significant level of 10%, 5%, and 1%, respectively.

Firstly, the second and third columns of [Table 4](#) list the regression results of [Equation \(2\)](#), that is, regress urban relative poverty incidence on education without introducing social capital. The results show that the coefficient of the years of education is negatively significant at the 1% level.

Secondly, the fourth column of [Table 4](#) reports the regression result of [Equation \(4\)](#), that is, the regression of social capital to the years of education of the family. It can be seen that the years of education have a positive impact on social capital, and its estimated coefficient is significant at the 1% level. This shows that families with higher education level will have more abundant social capital, and Hypothesis 2 holds. There are two main ways for education investment to produce social capital: first, going to school increases the number of classmates and teachers and enriches social networks; second, education can bring a kind of alumni identity and make it easier to develop mutual trust, especially for students who graduated from the same university. The regression results show that when other variables remain unchanged, the amount of social capital will increase by 5.7% for each additional year of education.

Finally, the fifth and sixth columns of [Table 4](#) show the regression results of formula (3), that is, regress urban relative poverty incidence on education while introducing social capital. The results show that the coefficients of the years of education and social capital are both significant at the 1% level. The marginal effect of social capital on the incidence of relative poverty is significantly negative, which indicates that the increase in social capital accumulation can improve urban relative poverty. Firstly, social capital can replace physical capital to a certain extent, providing financial security and emotional support for the relatively poor; second, the more developed the social network, the lower the asymmetry of employment information, so that the relatively poor have the opportunity to obtain higher paid jobs and escape the poverty trap.

Based on the above analysis, social capital is the mediating variable between education and urban relative poverty, and the mediating effect is a partial mediating effect. In other words, the impact of education on urban relative poverty has not only direct effects, but also indirect effects through social capital.

4.4. Heterogeneity analysis

The benchmark results and analysis of influence mechanisms show that education will increase social capital, and then the increase of social capital can significantly help urban households get rid of relative poverty. Is this influence mechanism regionally heterogeneous? This paper takes Confucian culture and the degree of marketization as starting points, and answers this question from a new research perspective. In order to protect the privacy of respondents, the CFPS database only provides respondents' provincial-level addresses and doesn't provide respondents' county-level addresses. Therefore, this paper further analyzes the heterogeneity of this mechanism at the provincial level based on CFPS2018.

4.4.1. Regional heterogeneity caused by Confucian culture

In Chinese society, the construction of interpersonal relationships and the expansion of social networks are strongly influenced by Confucian culture. The stronger the influence of Confucian culture, the more social capital may be generated in the process of education, thus resulting in regional heterogeneity. Referring to the research of Kung and Ma (2014), this paper uses the number of Confucian temples in each province as an indicator of the degree of Confucian cultural influence (Kung and Ma 2014)³. By consulting lists of national key protected cultural relic units (the first to the seventh batches) and the provincial-level protected cultural relic units in each province, the number of Confucian temples in each province was counted, as shown in Figure A1 in the Appendix.

There are 495 Confucian temples in China, with an average of about 15.97 per province. Taking this average as the boundary, this paper divides China's 31 provinces into two groups: provinces with a number of Confucian temples less than or equal to the average are classified as the group with weak Confucian cultural influence, and provinces with a number of Confucian temples greater than this average are classified as the group with strong Confucian cultural influence. Regression results show that Hypothesis 1, Hypothesis 2 and Hypothesis 3 are still established in either groups of strong or weak Confucian cultural influence. In a word, education can reduce urban relative poverty through the indirect way of improving social capital, regardless of strong or weak Confucian cultural influence. For the results of heterogeneity analysis based on the degree of Confucian cultural influence, see Table A1 in the Appendix.

In order to compare the mediating role of social capital in education poverty reduction under different Confucian culture influence degrees, this paper refers to Preacher (2007) and De Stavola et al. (2015), establishing the following mediating effect model:

$$Socapital = \alpha_0 + \alpha_1 Edulen + \alpha_2 Confucian + \alpha_3 Edulen * Confucian + \varepsilon_4 \quad (5)$$

$$\begin{aligned} Prob(Relpoverty = 1) = & \Phi(b_0 + b_1 Socapital + b_2 Edulen + b_3 Confucian \\ & + b_4 Edulen * Confucian + b_5 Socapita * Confucian + b_6 CV + \varepsilon_5) \end{aligned} \quad (6)$$

The variable *Confucian* represents the degree of influence of the Confucian culture. If the influence of the Confucian culture is weak, the value is 0, and if the influence of the Confucian culture is strong, the value is 1. According to the conditional indirect effect analysis method proposed by Preacher (2007), the conditional indirect effect is equal to $(b_1 + b_5 Confucian)(a_1 + a_3 Confucian)$, and the relative size of the mediation effect can be obtained by comparing the size of the conditional indirect effect when *Confucian* = 0 and *Confucian* = 1. The Bootstrap method is used for 500 cycles of calculation, and the results are shown in Table 5. It can be seen from Table 5 that the conditional indirect effect (0.006) when *Confucian* = 1 is greater than the conditional indirect effect (0.003) when *Confucian* = 0. This proves Hypothesis 4a of this paper, that is, the greater the influence of Confucian culture, the stronger the mediating role of social capital in poverty reduction through education. The results of the

Table 5. Conditional Indirect Effect under Different Confucian Cultural Influence Degree.

	CIE	Bootstrap SD	p value	95% CI
<i>Confucian</i> = 0	−0.003***	0.001	0.004	[−0.004, −0.001]
<i>Confucian</i> = 1	−0.006***	0.001	0.000	[−0.008, −0.003]

Notes: (1) "CIE" is the abbreviation of "Conditional Indirect Effect". (2) "SD" is the abbreviation of "Standard Deviation". (3) "CI" is the abbreviation of "Confidential Interval". (4) *, **, and *** indicate a significant level of 10%, 5%, and 1%, respectively.

heterogeneity analysis suggest that policy makers should pay more attention to the role of social capital in education poverty alleviation policies in areas with strong Confucian cultural influence, and fully expand the social network of poor families in schools to achieve better poverty reduction effects.

4.4.2. Regional heterogeneity caused by marketization

The role of social capital may be weakened by market forces. Different provinces have different degrees of marketization, resulting in regional heterogeneity. Following Sun et al. (2022), this paper uses the 2018 China Provincial Marketization Index published by the Beijing Institute of National Economics to measure the degree of marketization⁴. The data come from the China Market Index Database⁵. Taking the average of the marketization index as the boundary, this paper divides China's 31 provinces into two groups: provinces with a marketization index less than or equal to the average are classified as the group with a low degree of marketization, and provinces with a marketization index greater than this average are classified as the group with a high degree of marketization. For each province's Provincial Marketization Index, see Figure A2 in the Appendix. Regression results show that Hypothesis 1, Hypothesis 2 and Hypothesis 3 are still established in either groups of high or low marketization degree. In a word, education can reduce urban relative poverty through the indirect way of improving social capital regardless of high or low marketization degree. For the results of heterogeneity analysis based on the degree of marketization, see Table A2 in the Appendix.

In order to compare the mediating role of social capital in education poverty reduction under different marketization degrees, this paper refers to Preacher (2007) and De Stavola et al. (2015), establishing the following mediating effect model:

$$Socapital = \alpha_0 + \alpha_1 Edulen + \alpha_2 Market + \alpha_3 Edulen * Market + \varepsilon_4 \quad (5)$$

$$ProbRelapoverty = 1 = \Phi(b_0 + b_1 Socapital + b_2 Edulen + b_3 Market + b_4 Edulen * Market + b_5 Socapita * Market + b_6 CV + \varepsilon_5) \quad (6)$$

The variable *Market* represents the marketization degree. If the marketization degree is low, the value is 0, and if the marketization degree is high, the value is 1. According to the conditional indirect effect analysis method proposed by Preacher (2007), the conditional indirect effect is equal to $(b_1 + b_5 Market)(a_1 + a_3 Market)$, and the relative size of the mediation effect can be obtained by comparing the size of the conditional indirect effect when *Market* = 0 and *Market* = 1. The Bootstrap method is used for 500 cycles of calculation, and the results are shown in Table 6. It can be seen from

Table 6. Conditional Indirect Effect under Different Marketization Degree.

	CIE	Bootsrap SD	p value	95% CI
<i>Market</i> = 0	−0.008**	0.004	0.025	[−0.016, −0.001]
<i>Market</i> = 1	−0.006***	0.002	0.000	[−0.009, −0.003]

Notes: (1) “CIE” is the abbreviation of “Conditional Indirect Effect”. (2) “SD” is the abbreviation of “Standard Deviation”. (3) “CI” is the abbreviation of “Confidential Interval”. (4) *, **, and *** indicate a significant level of 10%, 5%, and 1%, respectively.

Table 6 that the conditional indirect effect (0.008) when *Market* = 1 is greater than the conditional indirect effect (0.006) when *Market* = 0. This proves Hypothesis 4b of this paper, that is, the higher the degree of marketization, the weaker the mediating role of social capital in poverty reduction through education. The results of the heterogeneity analysis suggest that policy makers should pay more attention to the role of social capital in education poverty alleviation policies in areas with low marketization degrees, and fully expand the social network of poor families in schools to achieve better poverty reduction effects.

4.5. Robustness checks

4.5.1. Alternative measurement of social capital

Referring to Baum et al. (2000), the frequency of communication between family and relatives, friends and neighbors is used as another measure of social capital to complete the robustness test (Baum et al. 2000). The results of robustness test are shown in **Table 7**.

It can be seen from the second and third columns of **Table 7** that, without adding the social capital variable, the family education years are still significantly negative at the level of 1%, and the marginal effect on the incidence of relative poverty is −0.035. The hypothesis 1 in this paper is still valid. Column 4 of **Table 7** shows that the number of years of family education has a positive impact on social capital at the significance level of 1%, and hypothesis 2 of this paper is still valid. It can be seen from columns 5 and 6 of **Table 7**, that after adding social capital variables, the coefficients of family education years and social capital are significant at the level of 1%. Based on the above results, it can be concluded that social capital is the intermediary

Table 7. Robustness Checks Results: Alternative measurement of social capital.

Dependent Variable	<i>Prob(Relapoverty = 1)</i>		<i>Socapital</i> Coefficients	<i>Prob(Relapoverty = 1)</i>	
	Coefficients	MEs		Coefficients	MEs
Household Education Level	−0.136*** (−21.206)	−0.035*** (−24.292)	0.027*** (7.930)	−0.134*** (−20.663)	−0.034*** (−23.550)
Social Capital	—	—	—	−0.129*** (−5.000)	−0.033*** (−5.046)
Control Variables	Included	Included	Included	Included	Included
Constant	−0.142 (−0.449)	—	3.025*** (21.442)	0.256 (0.788)	—
Observations	4976	4976	4976	4976	4976
Wald χ^2	2685*** (0.00)	—	—	2681*** (0.00)	—
Pseudo R^2	0.221	—	—	0.225	—
R^2	—	—	0.048	—	—

Notes: (1) “MEs” is the abbreviation of “Marginal Effects”. (2) Robust *t* statistic in parentheses, while *, **, and *** indicate a significant level of 10%, 5%, and 1%, respectively. (3) The table only shows marginal effects rather than estimated coefficients.

variable of education affecting urban relative poverty, and the intermediary effect is partial mediating effect. Hypothesis 3 of this paper is true.

4.5.2. Alternative measurement of the relative poverty line

For the purpose of robustness checks, this paper changes the relative poverty line to 50% of the median annual household income per capita. In addition, measure the family size directly by the number of family members may underestimate the annual household income per capita. There are two main reasons. Firstly, there are economies of scale in consumption within households. For example, Lazear and Michael (1980) found that if two single adults form a household, their consumption expenditure will be reduced by about 31% to 35%. Secondly, the consumption levels of family members of different ages are also very different. In order to exclude the influence of the above factors, this paper refers to the research of Wan et al. (2021) and uses the following formula to calculate the equivalent family size³¹:

$$h^e = h^{0.265-0.142*kid-0.130*old} \quad (5)$$

where h^e is the equivalent family size, h is the number of family members, kid represents the number of kids under 14 in the family, and old is the number of family members aged 60 years and older.

After adjusting the relative poverty line in the above way, the results of the robustness test are shown in Table 8. Due to space limitations, only marginal effects are reported in Table 8. When the control variables do not add social capital, no matter how the relative poverty line is set, the years of education are negatively significant at the level of 1%, indicating that Hypothesis 1 of this paper is established. Adjusting the relative poverty line does not affect Equation (4), so the coefficient of the explanatory variable X ($Edulen$) in Equation (4) is still significant at the 1% level, and Hypothesis 2 is still valid. When social capital is introduced to the control variables, no matter how the relative poverty line is set, the coefficients of the explanatory variable X ($Edulen$) and the mediating variable M ($Socapital$) are all significant at the 1% level, so the mediating effect test of social capital passes. Therefore, the Hypothesis 3 of this paper also holds.

Table 8. Robustness Checks Results (Marginal Effects).

Weakly Relative Poverty Line Control Variables	50% of the median annual household income per capita			
	Weighted by Equivalent Family Size		Unweighted	
	Exclude <i>Socapital</i>	Include <i>Socapital</i>	Exclude <i>Socapital</i>	Include <i>Socapital</i>
Household Education Level	−0.025*** (−23.484)	−0.024*** (−22.735)	−0.026*** (−23.895)	−0.025*** (−23.035)
Social Capital	—	−0.018*** (−11.656)	—	−0.020*** (−12.172)
Control Variables	Included	Included	Included	Included
Observations	6594	6594	6594	6594
Wald χ^2	1003.35*** (0.00)	1059.76*** (0.00)	1064.31*** (0.00)	1114.53*** (0.00)
Pseudo R^2	0.210	0.228	0.215	0.235

Notes: (1) Robust t statistic in parentheses, while *, **, and *** indicate a significant level of 10%, 5%, and 1%, respectively. (2) The table only shows marginal effects rather than estimated coefficients.

4.6. Endogeneity test results

The above benchmark model may have endogenous problems. On the one hand, relatively poor urban households may not be able to afford education costs, and correspondingly, the length of schooling is relatively short, resulting in a reverse causality problem. On the other hand, some unmeasured variables, such as the occurrence of major events, may not only make families become relatively poor, but also affect educational decision-making, resulting in omitted variable bias.

In order to solve these possible endogeneity problems, this paper uses the instrumental variable method for further analysis. ‘The average length of schooling of other families in the district’ will not only affect the family’s educational decision-making, but also have no direct impact on the family’s probability of relative poverty. Therefore, this paper uses ‘the average length of schooling of other families in the district’ as the instrumental variable.

In the first stage of regression, regress *Edulen* on the instrumental variables and the control variables mentioned above. In the second stage of regression, regress the predicted values of *Edulen* (i.e. $\widehat{Edulen}$) on *Prob(Relapoverty = 1)* and the control variables mentioned above. Results of two-stage least squares (2SLS) estimates are shown in Table 9. It can be seen that the results of the exogeneity test are significant at the 1% level, rejecting the null hypothesis that *Edulen* is exogenous. Results of the weak instrumental variable test are significant at the 5% level, rejecting the null hypothesis that *Edulen* is uncorrelated with the instrumental variable and confirming that the selected instrumental variable is not weak. The coefficient of instrumental variable is significantly positive at the level of 1%, and the coefficient of $\widehat{Edulen}$ is significantly negative at the level of 5%. Therefore, education can significantly reduce the possibility of urban households falling into relative poverty even if the endogenous problems that may exist are considered.

5. Conclusions and policy recommendations

5.1. Conclusions

Based on the data from CFPS2018 and from the perspective of social capital, this paper empirically studies the impact of education on the incidence of relative poverty in urban households. The valuable findings are as follows. Firstly, the increase in the number of years of education can help urban households get rid of relative poverty.

Table 9. Endogeneity Test Results.

Variables	First stage <i>Edulen</i>	Second stage <i>Prob(Relapoverty = 1)</i>
Average length of schooling of other families in the district	0.683*** (31.65)	—
$\widehat{Edulen}$	—	−0.028** (−2.08)
Control Variables		Included
Wald χ^2		215.6***
Exogeneity Wald Test		15.07***
Weak Instrumental Variable Wald Test		4.34**
Observations		6284

Notes: Robust *t* statistic in parentheses, while *, **, and *** indicate a significant level of 10%, 5%, and 1%, respectively.

Secondly, the poverty reduction effect of education has both a direct effect and a mediating effect by increasing social capital. Finally, the above-mentioned mechanisms show heterogeneity in different provinces: firstly, in regions with greater influence of Confucian culture, the mediating effect of social capital in education's role in poverty reduction is stronger; secondly, in regions with a higher degree of marketization, the mediating effect of social capital in education's role in poverty reduction is weaker.

5.2. Policy recommendations

Based on the conclusions, this paper puts forward the following targeted policy recommendations:

Firstly, governments at all levels should pay more attention to the role of education in the governance of urban relative poverty. Migrant workers and floating population in cities are more likely to fall into relative poverty, and they are disadvantaged in terms of living conditions, education, and health. Policy makers should provide them with opportunities to break through household registration restrictions through further education, and realize 'farmer-to-residential' and 'citizenization' opportunities, and help them integrate into urban society and get rid of relative poverty by improving their education level.

Secondly, governments should give full play to the role of social capital in education's role in poverty reduction. On college campuses, students receiving financial aid usually belong to poor families. The school should hold various activities to promote the interaction of such students with other students. In addition to arranging jobs, work-study centers should also organize students to participate in social activities, help children from relatively poor families integrate into urban society, and block the intergenerational transmission of poverty.

Thirdly, the degree of Confucian cultural influence and marketization in different regions should be fully considered when formulating education poverty alleviation policies. In provinces with a high degree of Confucian cultural influence and provinces with a low degree of marketization, the role of social capital should be emphasized in education poverty alleviation policies, and the social network of poor families should be fully expanded in schools, so as to better utilize the mediating effect of social capital between education and urban relative poverty alleviation.

Notes

1. The URL of the CFPS data platform is www.issp.pku.edu.cn/cfps/.
2. Following Williams(2012), the Stata command "margins" can be used to estimate marginal effects in the Probit model. Since the marginal effect of nonlinear models is generally not constant, there are different concepts of marginal effect. The marginal effect calculated in this paper is the "average marginal effect", that is, the marginal effect on each observation is calculated separately, and then the simple arithmetic average is performed.
3. Confucian temple number is adopted to proxy for the strength of Confucianism in the majority of the existing literature (eg., Kung and Ma, 2014; Chen et al., 2021; Zhang et al., 2021; etc.). The proxy variable is chosen on the reasoning as follows. Confucian temples are dedicated to Confucius, the great thinker and educator of ancient China. They are

associated with cultivating Confucius's thoughts and conducting cultural education. The more Confucian temples a province has, the more likely it is that Confucianism influences the residents of that province. In addition, the reason we do not use the number of temples per capital is answered in Kung and Ma (2014). They referred to the difference between Confucianism and other religions. The strength of other religions (such as Christianity) can be measured by, for instance, the share of communicants in a population, but the strength of Confucianism can not be measured in that way because "Confucianism resembles more of a set of social norms". The reason Kung and Ma (2014) chose total number rather than per-capita number of the Confucius temples is that the existence of a Confucian temple can establish Confucianism social norms, regardless of the population. As Kung and Ma (2014) said, "stronger Confucian norms would likely give rise to more Confucian temples being built, which would in turn deepen those norms".

4. China provincial marketization index is adopted to proxy for the marketization degree of each Chinese province in the majority of the existing literature (eg., Xin and Xin, 2017; Sun et al., 2022). The index measures the progress of transition towards market economy in 31 provincial administrative regions of the Chinese mainland. It is composed of five indexes with equal weights, and each index is composed of its subordinate indexes with equal weights. The five indexes are (1) the relationship between government and market, (2) the development of non-state-owned economy, (3) the development of product market, (4) the development of factor market, (5) the development of market intermediary organizations and the legal system environment. There are as many as 17 subordinate indexes of the five indexes, which is too long to describe here. For more details, please browse the website <https://cmi.ssap.com.cn/dataQuery> (in Chinese).
5. The website of China Market Index Database is <https://cmi.ssap.com.cn/>.

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Data availability statement

The data that support the findings of this study are available from the Institute of Social Science Survey (ISSS) of Peking University, China. Restrictions apply to the availability of these data, which were used under license for this study. Data are available at <http://www.issss.pku.edu.cn/cfps/> with the permission of ISSS.

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Appendix 1

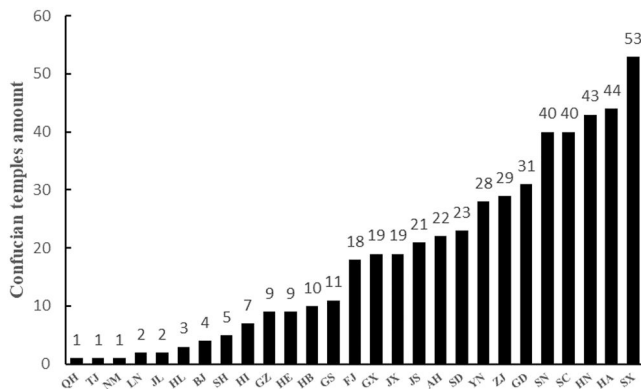


Figure A1. Number of Confucian Temples in each Province.

The bar chart shows the number of Confucian temples in every province in China. There are 495 Confucian temples in China, and the province with the largest number of Confucian temples is Shanxi Province, with 53.

Notes: (1) For simplicity, the provinces on the horizontal axis are represented by ISO 3166-2 codes and "CN-" is omitted. For example, the ISO 3166-2 code for Anhui Province in China is CN-AH, which is displayed as AH on the horizontal axis. (2) For provinces not listed in Figure A1, the number of Confucian temples is 0.

Data sources: lists of national key protected cultural relic units (the first to the seventh batches) and the provincial-level protected cultural relic units in each province.

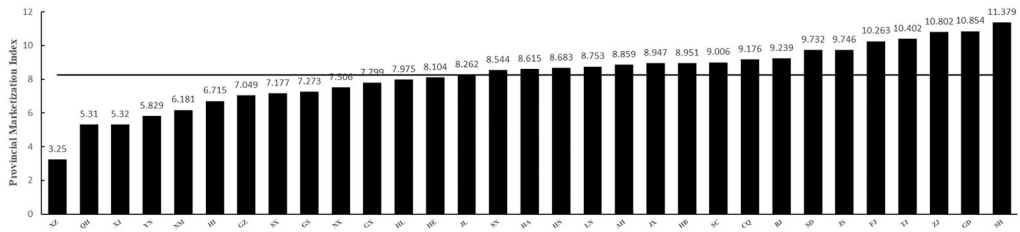


Figure A2. Provincial Marketization Index for each Province.

The bar chart shows Provincial Marketization Index for each province in China. The horizontal line shows the average of their marketization indexes. Shanghai, Guangdong and Zhejiang are the three provinces with the highest degree of marketization, which agrees with general knowledge.

Notes: For simplicity, the provinces on the horizontal axis are represented by ISO 3166-2 codes and "CN-" is omitted. For example, the ISO 3166-2 code for Anhui Province in China is CN-AH, which is displayed as AH on the horizontal axis.

Data sources: China Market Index Database whose website is <https://cmi.ssap.com.cn/>.

Table A1. Heterogeneity due to different confucian cultural influence degree.

Dependent Variable	Weak Confucian Cultural Influence			Strong Confucian Cultural Influence		
	<i>Socapital</i> Coefficients	<i>Prob(Relapoverty = 1)</i>		<i>Socapital</i> Coefficients	<i>Prob(Relapoverty = 1)</i>	
		Coefficients	MEs		Coefficients	MEs
Household Education Level	0.035*** (2.761)	−0.122*** (−16.454)	−0.027*** (−17.868)	0.073*** (5.922)	−0.095*** (−14.990)	−0.027*** (−16.260)
Social Capital	—	−0.073*** (−7.630)	−0.016*** (−7.778)	—	−0.076*** (−8.510)	−0.021*** (−8.738)
Control Variables	Included	Included	Included	Included	Included	Included
Constant	6.208*** (17.813)	0.556*** (2.864)	—	4.895*** (15.357)	0.104 (0.656)	—
Observations	3087	3087	3087	3507	3507	3507
Wald χ^2	—	463.37*** (0.00)	—	—	575.98*** (0.00)	—
Pseudo R^2	—	0.1767	—	—	0.164	—
R^2	0.069	—	—	0.086	—	—

Notes: (1) "MEs" is the abbreviation of "Marginal Effects". (2) Robust t statistic in parentheses, while *, **, and *** indicate a significant level of 10%, 5%, and 1%, respectively.

Table A2. Heterogeneity due to different marketization degree.

Dependent Variable	Low Marketization Degree			High Marketization Degree		
	<i>Socapital</i> Coefficients	<i>Prob(Relapoverty = 1)</i>		<i>Socapital</i> Coefficients	<i>Prob(Relapoverty = 1)</i>	
		Coefficients	MEs		Coefficients	MEs
Household Education Level	0.071*** (5.106)	−0.111*** (−12.169)	−0.034*** (−13.711)	0.053*** (4.927)	−0.107*** (−19.047)	−0.025*** (−20.471)
Social Capital	—	−0.091*** (−5.624)	−0.028*** (−5.838)	—	−0.071*** (−10.015)	−0.017*** (−10.210)
Control Variables	Included	Included	Included	Included	Included	Included
Constant	4.991*** (12.062)	0.770*** (3.251)	—	5.647*** (19.871)	0.051 (0.353)	—
Observations	1667	1667	1667	4927	4927	4927
Wald χ^2	—	284.74*** (0.00)	—	—	712.66*** (0.00)	—
Pseudo R^2	—	0.166	—	—	0.169	—
R^2	0.106	—	—	0.069	—	—

Notes: (1) "MEs" is the abbreviation of "Marginal Effects". (2) Robust t statistic in parentheses, while *, **, and *** indicate a significant level of 10%, 5%, and 1%, respectively.